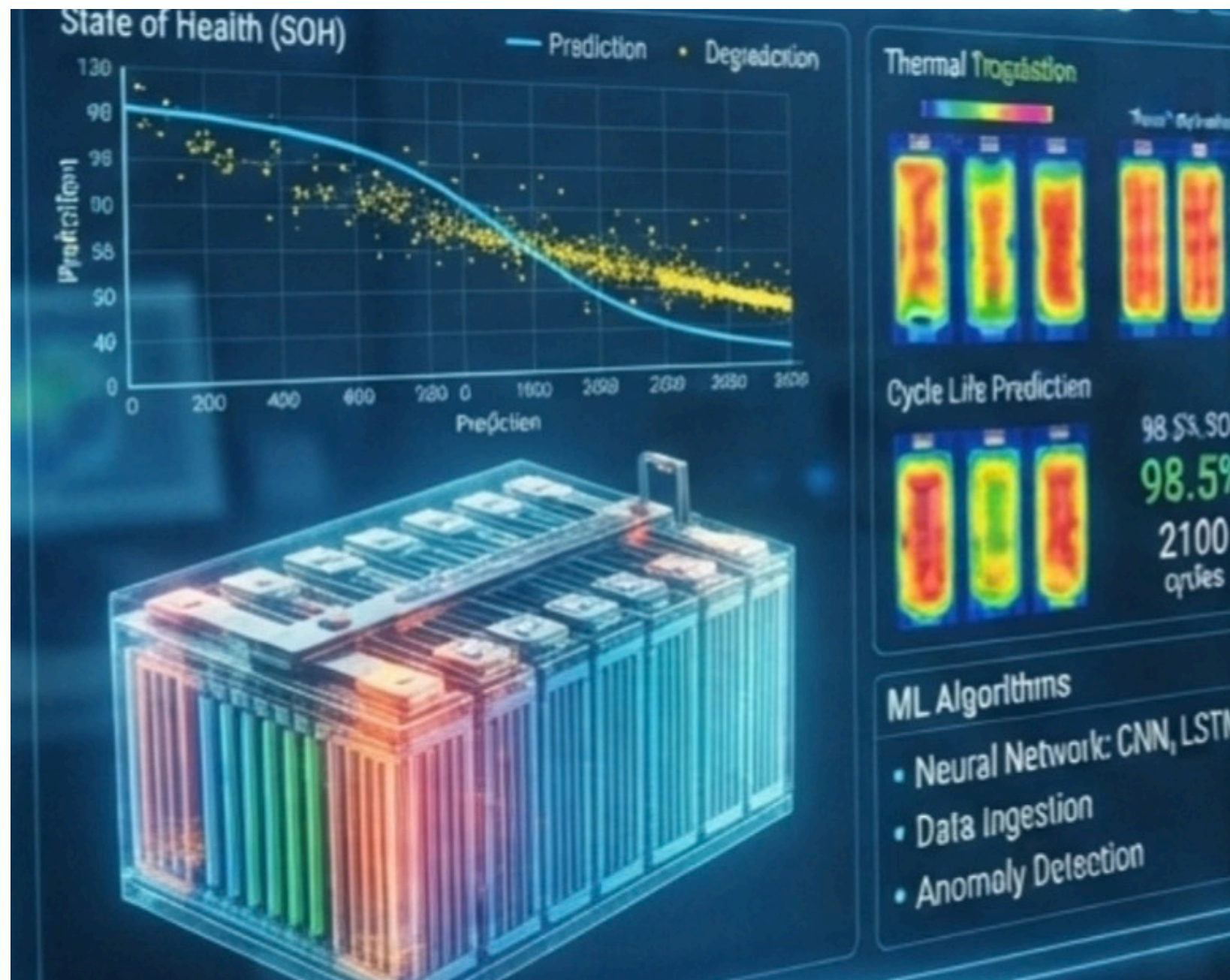




CrystalVolt

Machine Learning the Voltage of Electrode Materials in Metal-ion Batteries



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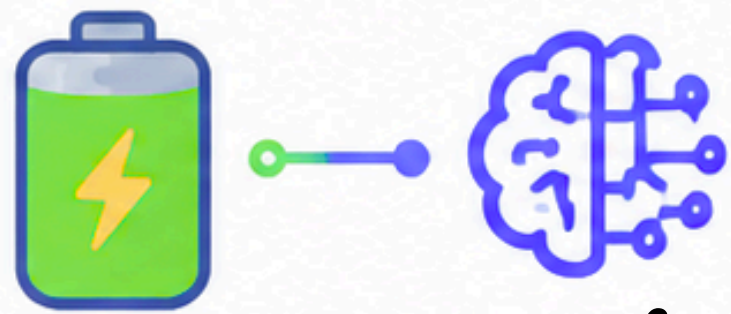
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Key Insights:

- The paper uses Machine Learning to predict the voltage of electrode materials.
- Material information such as chemical and elemental properties is converted into numerical features.
- ML models learn the relationship between these features and electrode voltage.

The Challenge:

- Finding the voltage of thousands of potential electrode materials is time-consuming and computationally expensive.
- The paper addresses this by using Machine Learning for fast voltage prediction.

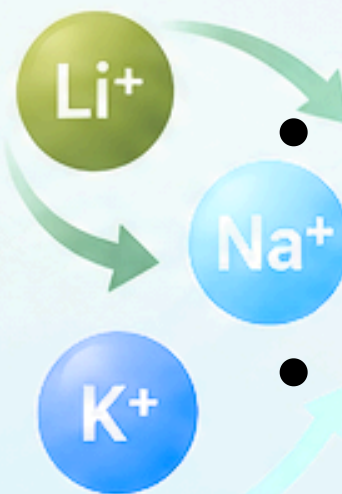


Importance of the Study:

- Reduces Computational cost of voltage prediction.
- Saves time compared with evaluating every material individually.
- Helps identify promising battery materials more efficiently.

The project aims to solve:

- Time-consuming voltage calculation of electrode materials.
- High computational cost of traditional methods such as DFT.
- Difficulty in screening thousands of potential electrode materials.
- Need for a faster method to predict electrode voltage.



In-Scope

- Material & crystal property representation
- Feature engineering for ML
- Average electrode voltage prediction
- ML model comparison and evaluation
- Prediction for unseen electrode materials
- Voltage-based material screening

Out-Of-Scope

- Experimental material testing
- New DFT calculations
- Battery manufacturing
- Cycle-life, safety and cost prediction
- Experimental validation of predicted materials



Our Proposed Solution

Our Approach:

Material Data Preprocessing

- Clean missing and duplicate data
- Standardize material information
- Prepare data for ML models

Feature Engineering

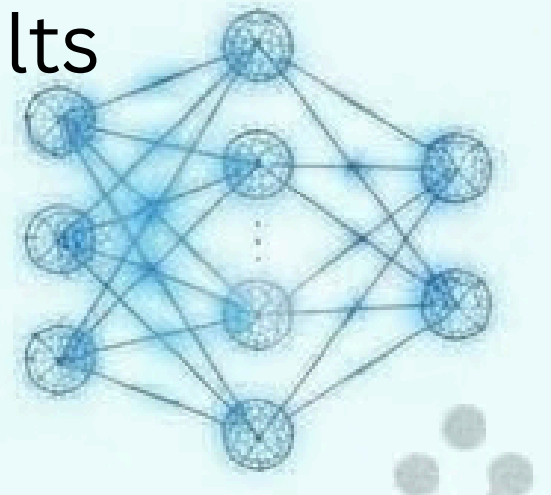
- Extract useful material features
- Convert features into numerical form
- Select important features for prediction

Machine Learning Models

- Random Forest → Robust prediction
- XGBoost → High accuracy
- Linear Regression → Baseline model

Voltage prediction

- Predict average electrode voltage
- Use material features as input
- Display predicted voltage in volts (V)



Expected Outcome:

Accurate prediction of electrode voltage

- Predict electrode voltage accurately
- Compare predicted and actual voltage
- Evaluate model accuracy

Fast Prediction for new materials

- Enter new material details
- Get voltage prediction quickly
- Reduce testing time and cost

Efficient screening of potential electrode materials

- Quickly screen new materials
- Identify promising electrode materials
- Reduce experimental testing



Machine Learning the Voltage of Electrode Materials in Metal-ion Batteries

Complete Project Architecture – From Data to Prediction

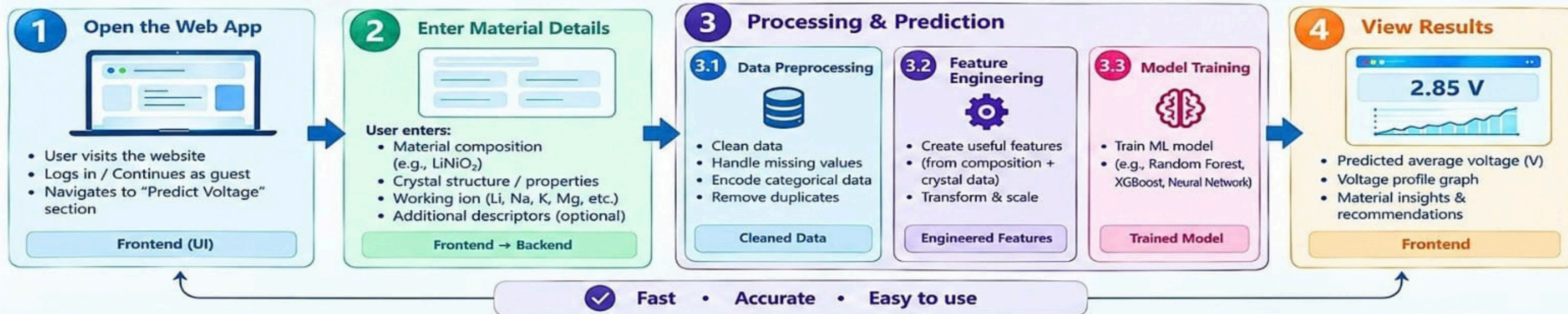


Smarter Materials → Better Batteries → A Greener Future



1. How the Project Works (User Flow)

When the user wants to see the result



Example – What the User Sees

Input Form

Material Composition

Crystal Structure

Working Ion

Additional Descriptor (optional)

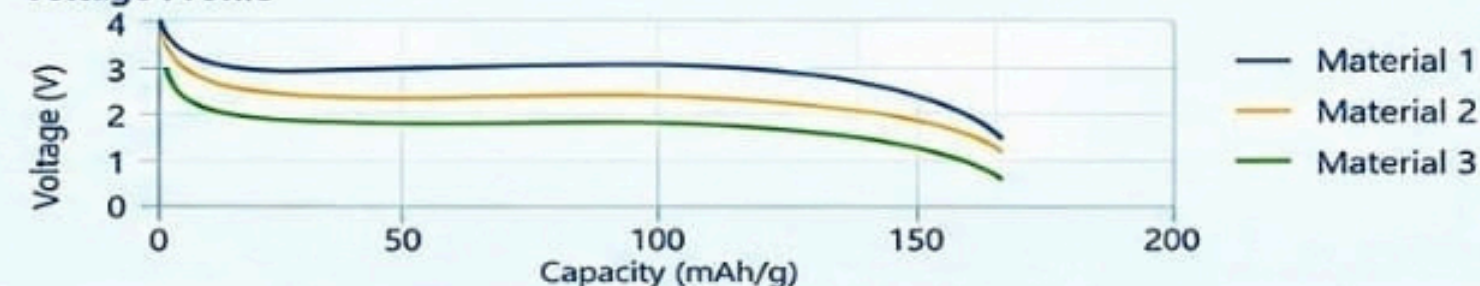
Predict Voltage

Result Page

Predicted Average Voltage

2.85 V

Voltage Profile



Additional Information

- Crystal structure: Layered
- Space group: R-3m
- Working ion: Li
- Predicted stability: High



This material shows good voltage performance and is a potential candidate for metal-ion batteries.



Models Used



Random Forest
(ensemble learning)



XGBoost
(gradient boosting)



Neural Network (MLP)
(deep learning)

Conclusion:

PROGRESS:

- Defined the ML problem and objective
- Analyzed the battery-voltage dataset
- Identified features and target
- Selected the ML models

NEXT MILESTONE:

- Complete data preprocessing & scaling
- Train ML models
- Compare models using MAE, RMSE & R^2
- Optimize the best model
- Build the prediction interface

THANK YOU!

